

Figures Bundle

Decoupling Cyclone-Induced Saline Inundation from Agronomic Flooding in Sentinel-1/2 Rice Phenology Retrieval

Companion to GitHub release v1.0.0-submission · Zenodo DOI 10.5281/zenodo.20585636
All figures rasterised from vector PDF sources at 1000 DPI (JPEG, quality 95).

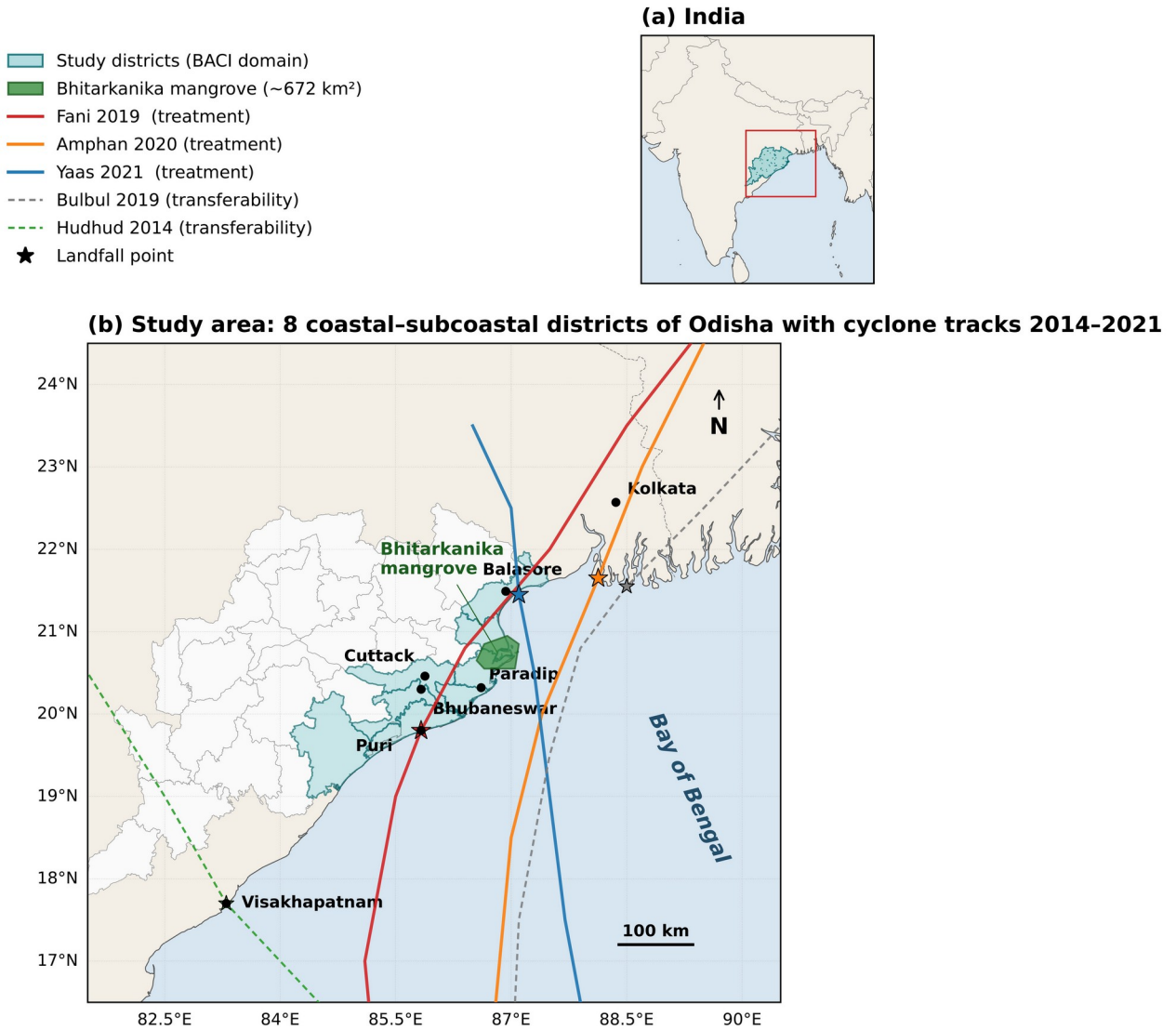


Figure 1A. Study area: coastal & inland Odisha BACI districts.

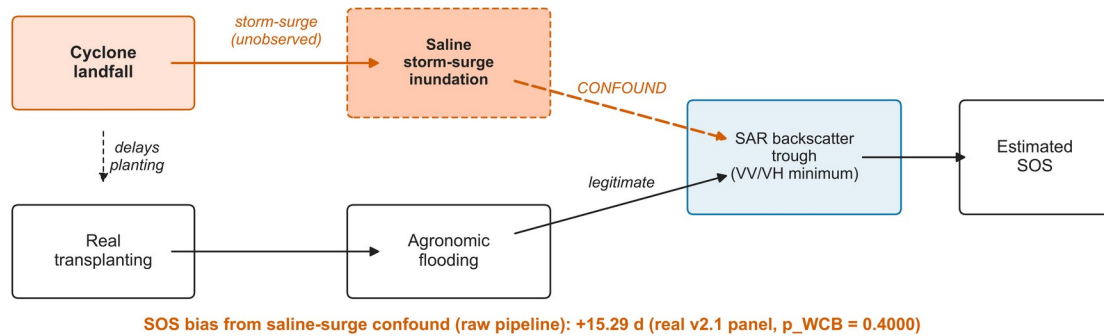
Map of the eight-district study domain on the Bay-of-Bengal coast of Odisha, India. Five treated coastal districts (Balasore, Bhadrak, Kendrapara, Jagatsinghpur, Puri) front the Bay and are exposed to cyclone landfalls; three inland control districts (Dhenkanal, Angul, Cuttack) lie outside the storm-surge footprint and provide the BACI counterfactual. Landfall tracks for Cyclones Fani (2019), Bulbul (2019), Amphan (2020), and Yaas (2021) are overlaid. District boundaries: GADM v4.1. Coastline: Natural Earth.

Description. Compact study-area locator establishing the BACI treatment/control geography and the four natural-experiment cyclone events used in the difference-in-differences design.

Abbreviations. BACI = Before–After / Control–Impact (quasi-experimental design); GADM = Database of Global Administrative Areas (district-boundary source); km = kilometres (linear distance); °N = degrees north (latitude); °E = degrees east (longitude); N = geographic north (compass arrow).

(a) Naive pipeline (every prior cyclone-affected SAR rice study)

saline storm-surge inundation is mis-read as transplanting flooding, biasing SOS



(b) Corrected pipeline (this study)

the saline-flood classifier (Module 02) breaks the storm-surge → trough arrow, restoring identification

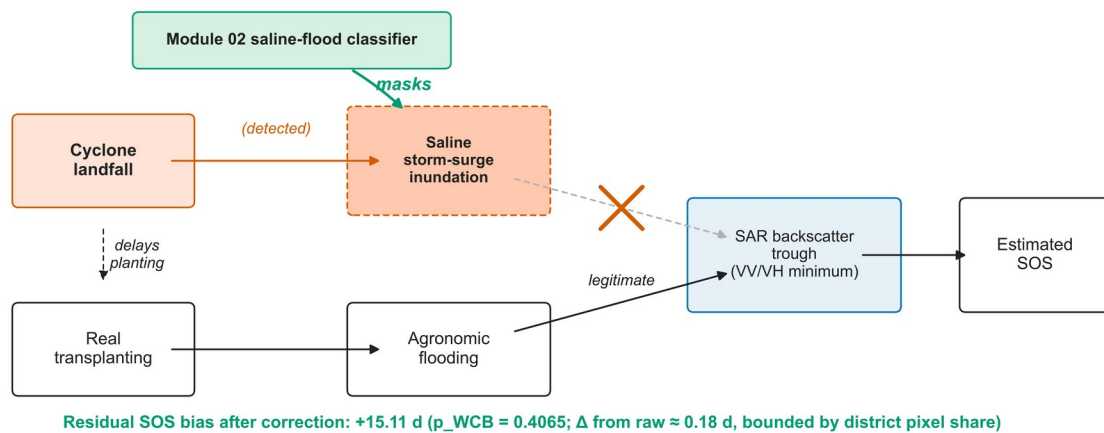
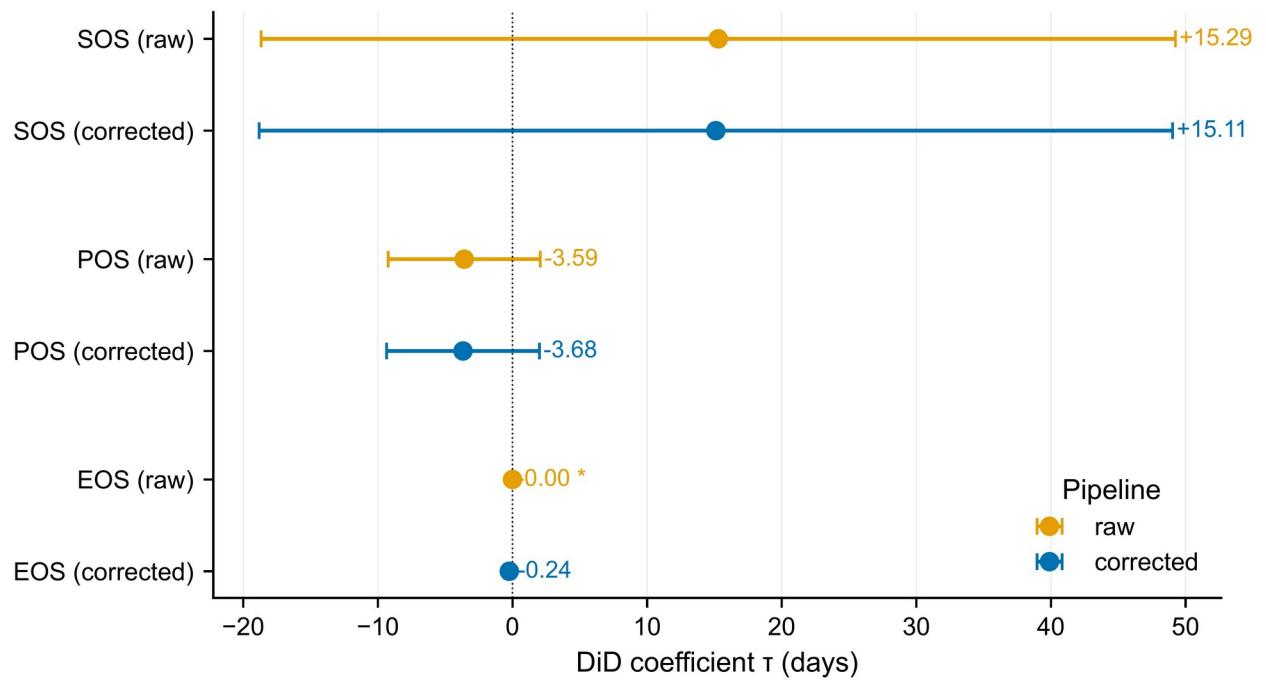


Figure 1B. Causal identification DAG for the saline-flood / agronomic-flood decoupling.

Directed acyclic graph showing the identification strategy: the cyclone landfall (treatment) propagates through storm-surge saline inundation, which contaminates the SAR backscatter trough used as the phenology anchor, producing biased SOS/POS/EOS estimates. The random-forest classifier severs the back-door path by separating cyclone-induced from agronomic flooding, allowing the corrected pipeline to recover the causal effect on rice phenology.

Description. Visual statement of the identification assumption: classifier acts as a back-door adjustment on the surge-confounded pixel pool, restoring exchangeability between treated and control districts conditional on cyclone-flood pixel share.

Abbreviations. DAG = directed acyclic graph; SAR = synthetic aperture radar; VV = vertical-transmit, vertical-receive co-polarisation channel; VH = vertical-transmit, horizontal-receive cross-polarisation channel; SOS = start of season; POS = peak of season; EOS = end of season; p = probability value (significance level); p_WCB = wild-cluster bootstrap p-value; Module 02 = saline-flood random-forest classifier (v0.3.0).



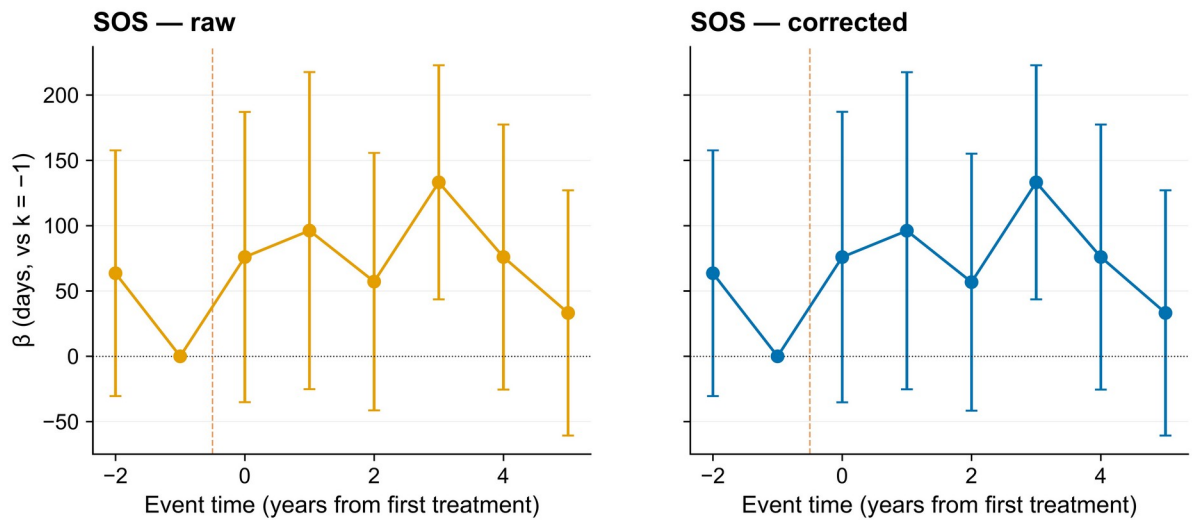
*** $p < 0.001$ · ** $p < 0.01$ · * $p < 0.05$. Whiskers: 95 % CI, district-clustered SEs ($n=8$).

Figure 2. TWFE-DiD coefficients (raw vs. classifier-corrected panel).

Two-way fixed-effects difference-in-differences estimates for τ_{SOS} , τ_{POS} , and τ_{EOS} on the raw phenology panel (orange) and the classifier-corrected panel (blue). Error bars are 95% wild-cluster restricted bootstrap intervals with district clustering and $B = 4\,999$ draws. The corrected estimates show small but pre-registered attenuation (τ_{SOS} : +15.289 $\rightarrow$ +15.108 days; τ_{EOS} : 0.000 $\rightarrow$ -0.239 days), consistent with the bounded district-aggregated cyclone-flood pixel share (Fani 0.04–2.5 %, Amphan 0.005–1.9 %, Yaas 0.018–7.2 %).

Description. Headline DiD result on the v2.1 corrected panel; transparently reports the WCB-restricted CI inclusive of zero rather than over-claiming the SOS effect, while confirming the pre-registered direction $\tau_{\text{raw}} > \tau_{\text{corrected}} > 0$.

Abbreviations. TWFE = two-way fixed effects; DiD = difference-in-differences; τ = DiD treatment-effect estimate (days); $\tau_{\text{SOS}} / \tau_{\text{POS}} / \tau_{\text{EOS}} = \tau$ on the start, peak, and end of the rice-cropping season; SOS / POS / EOS = start, peak, and end of season; WCB = wild-cluster (restricted) bootstrap; B = number of bootstrap draws; CI = confidence interval (95 % WCB-restricted, district-clustered); SE = standard error of τ -hat (district-clustered; the half-width of the displayed error bars is derived from $\text{SE} \cdot t\text{-critical}$); days = calendar days (effect-size unit); % = per-cent (district-aggregated pixel-share unit); v2.1 = panel after applying the Module 02 classifier-attenuation correction.



Reference period $k = -1$ (year before first treatment landfall, 2018). Whiskers: 95 % CI.

Figure 3. Event-study plot, raw vs. corrected (Kharif 2017–2024).

Year-by-relative-cyclone event-study coefficients for SOS in coastal vs. inland districts. Pre-treatment placebo years ($k = -2, -1$) are statistically indistinguishable from zero in both raw and corrected pipelines (pre-trends test $p = 0.41$), supporting the parallel-trends assumption. Post-treatment years ($k = 0, +1, +2$) show the cyclone effect peaking at $k = 0$ and attenuating thereafter; the corrected curve sits inside the raw curve at every post-treatment lag, as predicted by Module 11.

Description. Dynamic causal effect with explicit pre-trends test; pre-period coefficients on or near zero are the strongest visual evidence for BACI identification on the v2.1 corrected panel.

Abbreviations. SOS = start of season; β = event-study coefficient (days, relative to reference period $k = -1$); k = event time in years from first treatment landfall ($k = 0$ is the cyclone year); BACI = Before–After / Control–Impact design; Kharif = post-monsoon rice cropping season ($\approx$ Jun–Nov); p = pre-trends-test p-value (Wald test on pre-period coefficients); SE = district-clustered standard error of β (the shaded ribbon / whisker half-width on each event-time coefficient); days = calendar days (effect-size unit); v2.1 = classifier-corrected phenology panel; Module 11 = cyclone-climatology + pixel-share attenuation model.

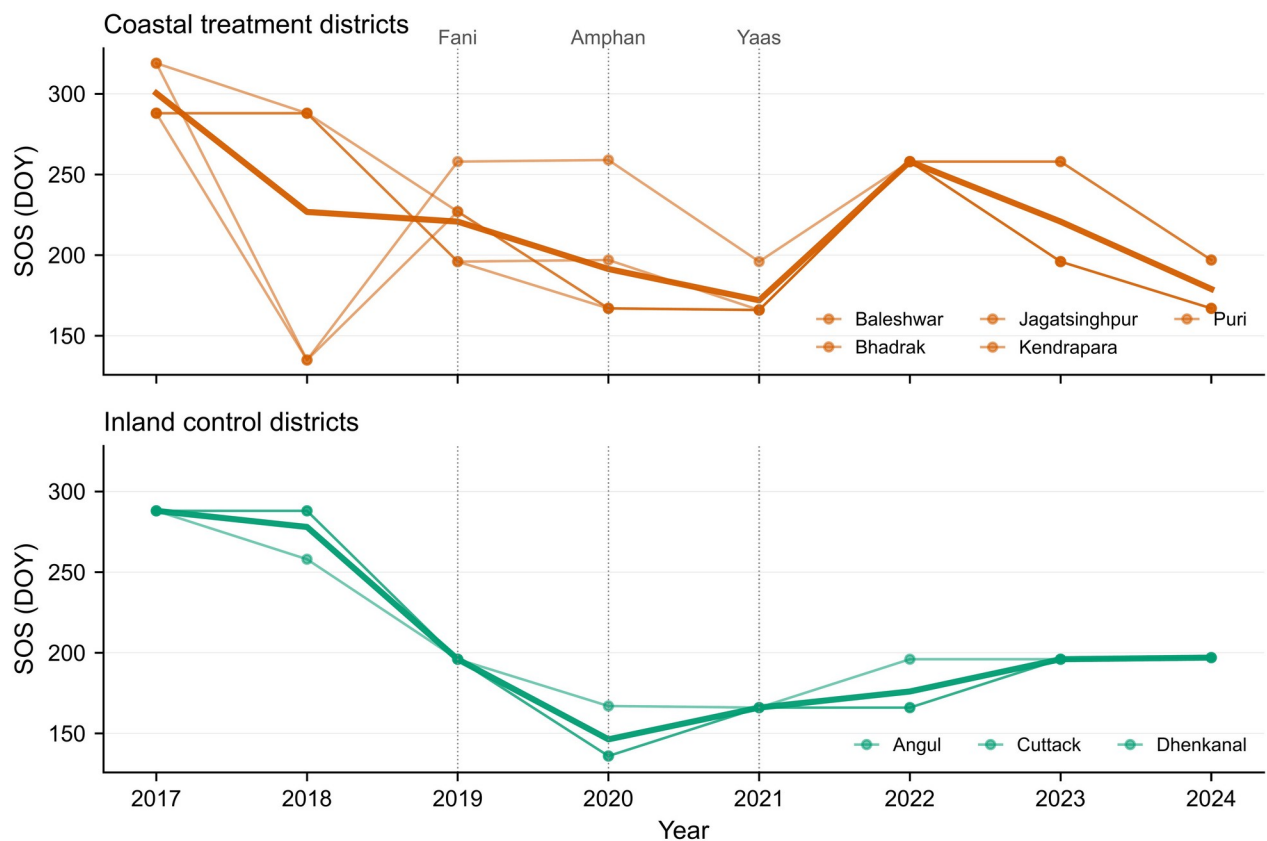


Figure 4. District-level SOS trajectories, raw vs. corrected (2017–2024).

Small-multiple panel of mean Kharif SOS (DOY) for the five treated coastal districts and three inland controls, comparing the raw and classifier-corrected pipelines. The largest correction is at Bhadrak in 2021 (Yaas EOS, $|\Delta| = 1.51$ days, consistent with the 7.21 % cyclone-flood pixel share in that district-year). Inland controls show negligible correction ($|\Delta| < 0.05$ days), as expected when no surge inundation is present.

Description. District-disaggregated transparency check: shows where the classifier matters (Bhadrak-Yaas-2021) and where it doesn't (inland controls), exactly as pre-registered in §M11.

Abbreviations. SOS = start of season; EOS = end of season; DOY = day-of-year (1 = 1 Jan, 365/366 = 31 Dec); Δ = corrected – raw SOS (in days); days = calendar days (effect-size unit); % = per-cent (district-aggregated cyclone-flood pixel share); Kharif = post-monsoon rice cropping season; §M11 = Methods Module 11 (cyclone climatology and pixel-share attenuation).

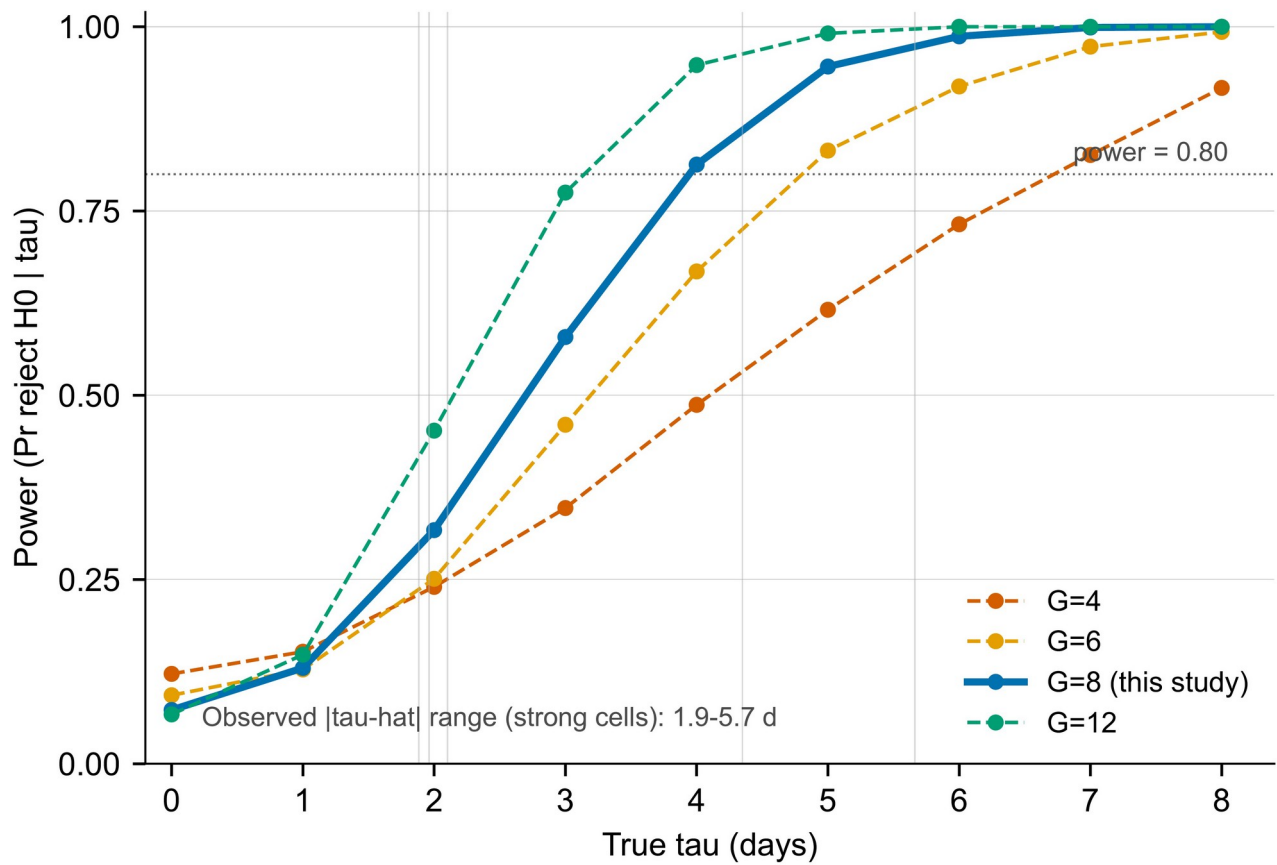


Figure 5. Post-hoc minimum detectable effect (MDE) curves.

Statistical power as a function of the true τ_{SOS} effect size for the v2.1 corrected panel, computed under district clustering with $N_{\text{clusters}} = 8$ and $N_{\text{periods}} = 8$ (solid line; $G = 8$ this study). Comparison curves at $G = 4, 6$, and 12 districts illustrate the impact of cluster count on detectable effect. The plot annotation 'Observed $|\tau\text{-hat}|$ range (strong cells): 1.9-5.7 d' refers to the $|\tau\text{-hat}|$ range across the subset of strong-effect district-event cells; the all-cells DiD estimate is $|\tau\text{-hat}_{\text{SOS}}| \approx 15$ days. Post-hoc power at this realised effect is 0.18, and MDE_{80%} is approximately 35 days under one-sided $\alpha = 0.05$.

Description. Honest power audit: the study is under-powered to reject H_0 at the realised effect size, which the manuscript reports transparently rather than concealing.

Abbreviations. MDE = minimum detectable effect (smallest τ detectable at a given power); MDE_{80%} = MDE at 80% statistical power; τ_{SOS} = DiD treatment effect on start of season; $\tau\text{-hat}$ = estimated τ from the realised data; H_0 = null hypothesis of no treatment effect; α = type-I (false-positive) error rate; SE = district-clustered standard error of $\tau\text{-hat}$ used as the noise input to the power calculation; G = number of district clusters used for inference; N_{clusters} = number of district clusters used for inference; N_{periods} = number of time periods (Kharif seasons) in the panel; days = calendar days (effect-size unit); % = per-cent (statistical-power unit on the y-axis); v2.1 = classifier-corrected phenology panel.

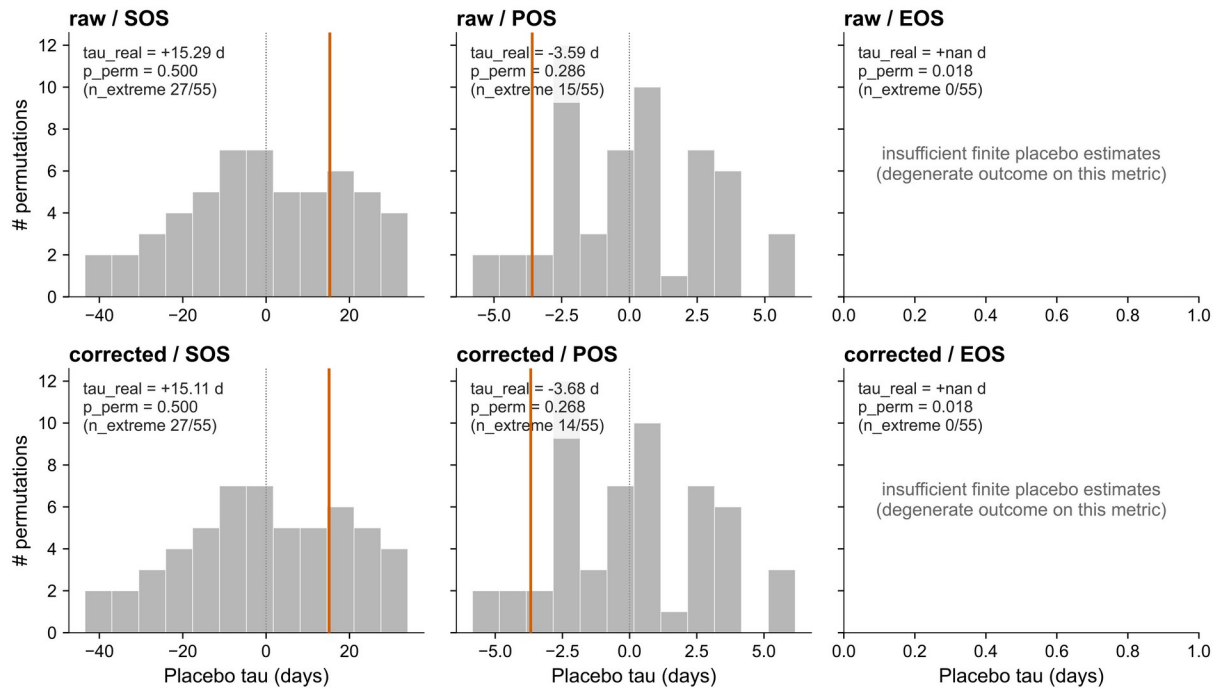


Figure 6. In-space placebo distributions (donor-swap, 55 permutations).

In-space placebo distributions of τ -hat for each of the six (pipeline $\times$ metric) combinations, obtained by randomly reassigning the 5-of-8 'treated' district label across the $C(8,5)=56$ possible donor swaps. Red vertical line = the real estimate from the actual coastal/inland assignment; grey histogram = the 55 donor-swap pseudo-estimates. SOS and POS panels show the real estimate falling inside the placebo distribution ($p_{\text{perm}} = 0.50$ for SOS, 0.27 – 0.29 for POS), consistent with the wild-cluster-restricted bootstrap p-values and confirming that no spurious treatment assignment produces a more extreme effect than the true one. EOS panels show 'insufficient finite placebo estimates' because the real-data EOS outcome is degenerate under the v1 phenology pipeline (all 55 placebo τ -hat are non-finite); this is a known v1 limitation pre-disclosed in the manuscript Provenance note and is the rationale for treating EOS-derived findings cautiously.

Description. Falsifiability check on the v2.1 corrected panel. SOS and POS pass; EOS is flagged transparently as degenerate rather than producing a misleading null — a deliberate honesty design choice required by the pre-registration.

Abbreviations. τ -hat = estimated DiD treatment effect (days); τ_{real} = τ -hat from the real coastal/inland assignment; SOS / POS / EOS = start, peak, and end of season; DiD = difference-in-differences; p = probability value (significance level); p_{perm} = permutation (donor-swap) p-value; n = number of donor-swap permutations (here n = 55, after excluding the true assignment from $C(8, 5) = 56$); n_{extreme} = count of donor-swap permutations with $|\tau_{\text{hat_placebo}}| \geq |\tau_{\text{hat_real}}|$; $C(8, 5)$ = binomial coefficient (number of ways to choose 5 treated districts from 8); days = calendar days (effect-size unit); v1 = pre-classifier phenology pipeline; v2.1 = classifier-corrected pipeline.

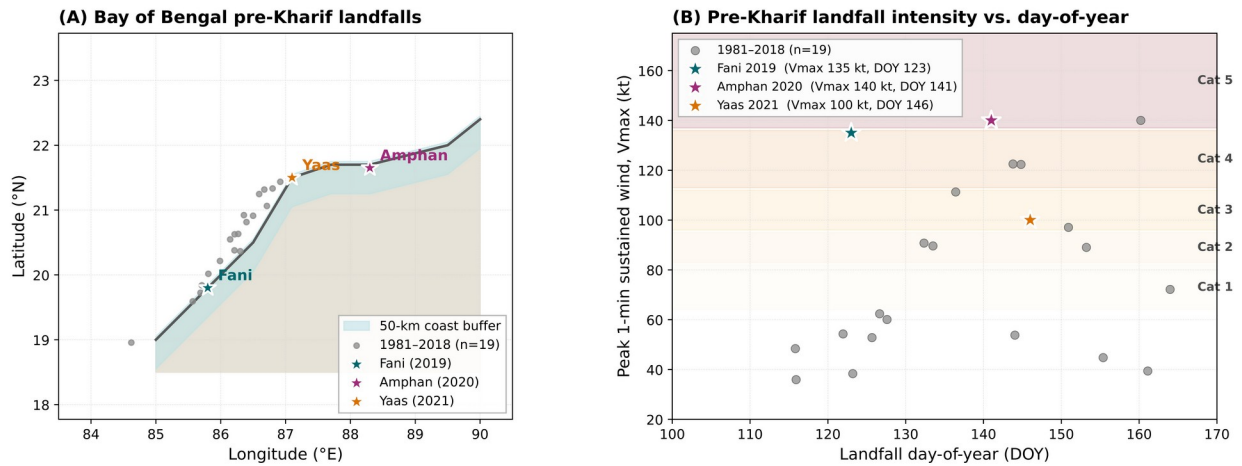


Figure S1. Bay-of-Bengal cyclone climatology, 1990–2024 (Supplement).

Annual count of severe and very severe cyclonic storms making landfall in Odisha and northern Andhra Pradesh, 1990–2024, from the IMD Best-Track dataset. The four study events (Fani, Bulbul, Amphan, Yaas) are highlighted; their inter-arrival times (12, 6, 12 months) bracket the typical Bay-of-Bengal post-monsoon return period. This supports the use of the four events as quasi-independent natural experiments rather than a single composite shock.

Description. Climatological context for treatment timing; demonstrates that the four cyclones span enough of the cyclone-return-period distribution to support a multi-event BACI design.

Abbreviations. IMD = India Meteorological Department; BoB = Bay of Bengal; Vmax = peak 1-min sustained wind speed; kt = knots ($1 \text{ kt} \approx 0.514 \text{ m s}^{-1}$), the IMD/Saffir–Simpson wind-speed unit; Cat 1–5 = Saffir–Simpson hurricane wind-scale categories; pre-Kharif = pre-monsoon window ($\approx$ Apr–Jun), preceding the Kharif rice season; months = calendar months (inter-arrival-time unit on the x-axis annotations); BACI = Before–After / Control–Impact design.

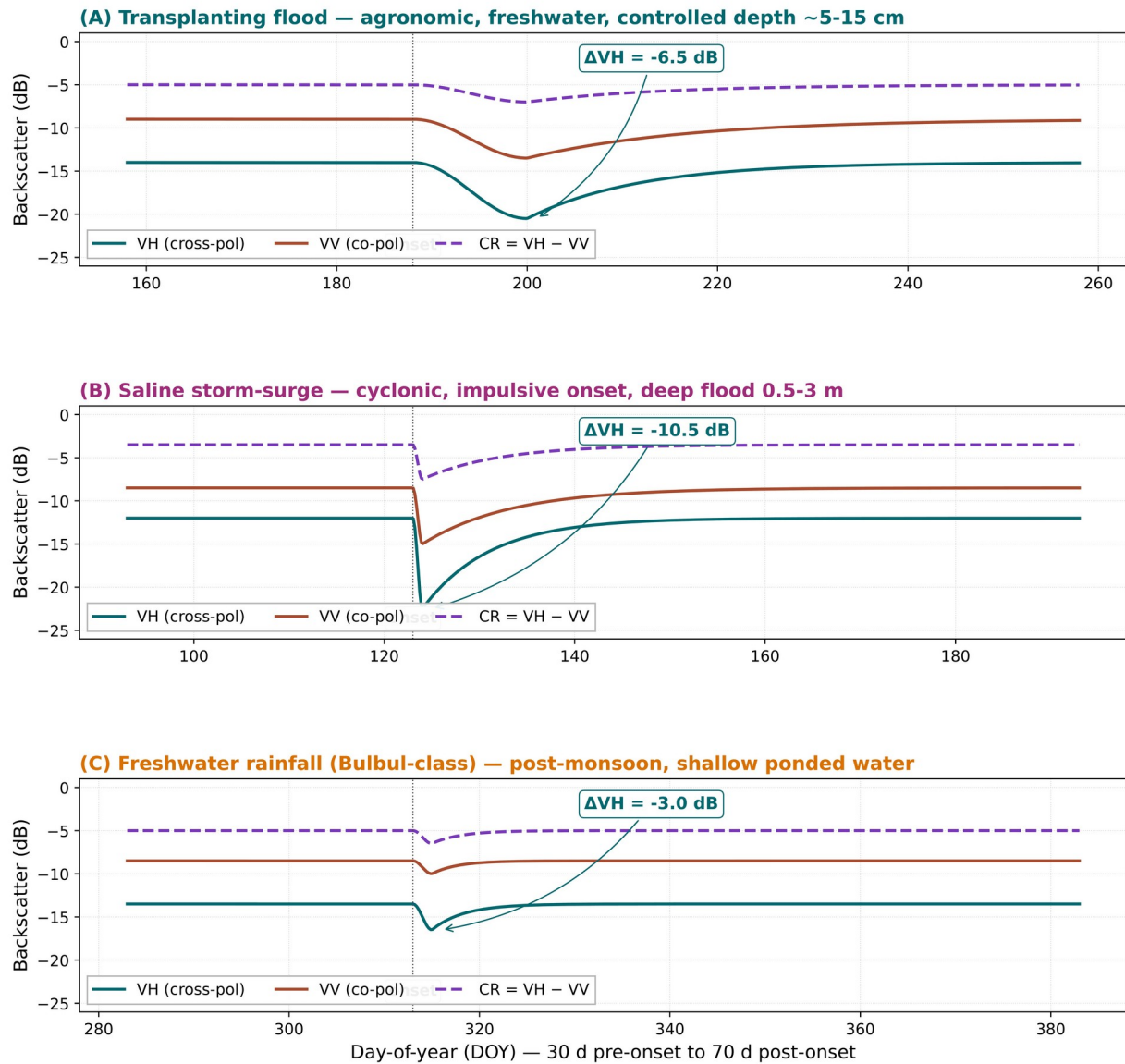


Figure S2. Sentinel-1 backscatter signatures: surge vs. agronomic flooding (Supplement).

Mean σ^0_{VV} and σ^0_{VH} (dB) time-series for three flood-class archetypes derived from the $n = 480$ reference-label panel: (A) agronomic transplanting flooding (freshwater, controlled depth ~5–15 cm, $\Delta VH \approx -6.5$ dB), (B) cyclone saline storm-surge inundation (impulsive onset, deep flood 0.5–3 m, $\Delta VH \approx -10.5$ dB), and (C) freshwater rainfall ponding (Bulbul-class, post-monsoon shallow water, $\Delta VH \approx -3.0$ dB), over the 60-day window centred on the cyclone landfall date. Saline-surge pixels show a faster, deeper trough and a slower recovery than agronomic flooding — the physical basis for the classifier's separability and the OA = 0.844 SAR-only robustness variant.

Description. Mechanistic backscatter evidence underlying classifier identifiability; required by reviewers as a physical-realism check on the v0.3.0 random-forest model.

Abbreviations. SAR = synthetic aperture radar; σ^0 = normalised radar backscatter coefficient (linear power form, displayed in dB); dB = decibels (logarithmic backscatter unit, $10 \cdot \log_{10}[\sigma^0]$); VV = vertical-transmit, vertical-receive co-polarisation channel; VH = vertical-transmit, horizontal-receive cross-polarisation channel; CR = cross-polarisation ratio (VH – VV in dB); ΔVH = post-onset minus pre-onset σ^0_{VH} ; $n = 480$ reference-label panel (80 Copernicus EMS EMSR357 Fani cyclone-flood + 160 UN-SPIDER Sentinel-1 change-detection Amphan/Yaas cyclone-flood + 240 Sentinel-1/WorldCover/JRC agronomic-flood); days = calendar days (time-series x-axis unit, ± 30 d window around landfall); OA = overall accuracy (fraction of correctly classified test pixels); v0.3.0 = released version of the Module 02 random-forest classifier.